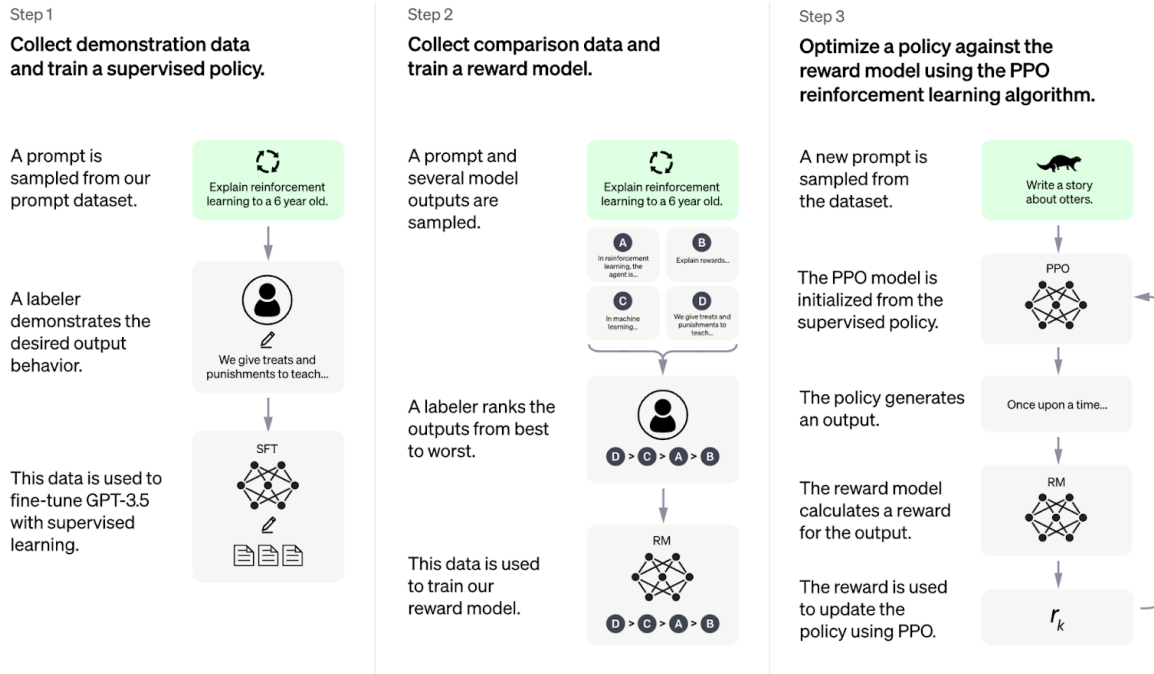


Diagram of RLHF (<https://openai.com/blog/chatgpt>)



One limitation of RLHF: humans make mistakes ([Casper et al. 2023](#))

Tractable: Humans make simple mistakes due to limited time, attention, or care. Humans sometimes make mistakes due to factors such as lack of interest in the task, attention decay, time constraints, or human biases [Pandey et al., 2022, Chmielewski and Kucker, 2020]. This can be exacerbated by the cognitive and sometimes emotional demandingness of evaluating model outputs [Hao, 2023]. Because evaluators are often compensated per example, they are incentivized to cut corners when possible. Mistakes can be correlated across annotators. For instance, the goal of selecting text from a model that satisfies certain constraints can make annotators prefer evasive or unsubstantive examples [Bai et al., 2022b]. Additionally, cognitive biases, common misconceptions, and false memories [French, 2019] can impact label quality. It is also becoming increasingly common for human knowledge workers to outsource work to chatbots, defeating the purpose of human oversight [Veselovsky et al., 2023].

Empirical scaling laws across compute, data, and parameters ([Kaplan et al. 2020](#))

